# Javad Maheri

Information Theory • Distributed Computing & Learning • Learning Theory • Combinatorics

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LinkedIn • GitHub • Google Scholar • EURECOM profile

## Research Profile

PhD, Sorbonne Université / EURECOM (awarded September 2026). I characterize **exact communication limits of distributed computing and learning**: which data each worker should hold so that any requested set of computations is served with minimum communication. My results pair **information-theoretic converse bounds** with explicit, task-blind constructions that meet them, and extend to **random sampling of the requested computations** through concentration and high-dimensional probability. My next step is to bring this toolkit to **large-model training**: the communication and load-balancing tradeoff of **mixture-of-experts** models, where the router samples the requested computations at every step, together with **finite-sample guarantees** for placements learned from observed routing data.

## Education

**Ph.D., Communication Systems** — Sorbonne Université / EURECOM, Biot, France 2023 – 2026
Advisor: Prof. Petros Elia. Degree awarded September 2026. Thesis: *Universally Optimal Combinatorial Schemes for Distributed Computing and Learning*. Jury: Prof. Dirk Slock (EURECOM), Prof. Michèle Wigger (CentraleSupélec), Prof. Giuseppe Caire (TU Berlin), Prof. Sheng Yang (CentraleSupélec).

**Graduate Coursework, M.Sc. Program in Electrical Engineering (Communication Systems)** — 2021 – 2022
Sharif University of Technology, Tehran, Iran
Graduate GPA **19.04/20** (cohort average 16.37). Courses listed under Relevant Coursework below.

**B.Sc., Electrical Engineering & Mathematics (Minor in Math)** — Sharif University of Technology, 2016 – 2021
Tehran, Iran
Graduated with high honors. Ranked **7th among 180,000+ candidates** in the Iranian national university entrance examination.

## Relevant Coursework (Sharif University of Technology, grades out of 20)

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|---|---|
| **Information & communication** | Information Theory and Coding 19.4 • Special Topics in Communications 20.0 • Adaptive Filters 18.9 |
| **Probability & statistics** | Stochastic Processes 20.0 • Engineering Probability & Statistics 19.5 |
| **Learning** | Theory of Machine Learning 20.0 • Deep Learning 19.1 • Tensor Decompositions 19.4 • Compressed Sensing 17.9 |
| **Optimization** | Convex Optimization II 19.2 • Convex Optimization I 18.5 • Numerical Optimization Methods 18.7 |
| **Analysis & discrete math** | Real Analysis I 20.0 • Real Analysis II 20.0 • Engineering Mathematics 19.9 • Graph Theory 17.0 |

## Publications

- **J. Maheri**, K. K. Krishnan Namboodiri, P. Elia. *Universal and Asymptotically Optimal Data and Task Allocation in Distributed Computing. IEEE Transactions on Information Theory* (Accepted), 2026.
- **J. Maheri**, K. K. Krishnan Namboodiri, P. Elia. *Fundamental Limits of Hypergraph Edge Partitioning under Independent Edge Sampling.* Submitted to *IEEE Transactions on Information Theory*, 2026.
- **J. Maheri**, K. K. Krishnan Namboodiri, P. Elia. *Order-Optimal Task Allocation in Distributed Computing via Interweaved Cliques.* IEEE Int. Symposium on Information Theory (ISIT), 2026.
- **J. Maheri**, P. Elia. *Constructing Hamiltonian Decompositions of Complete $k$-Uniform Hypergraphs.* IEEE Int. Symposium on Information Theory (ISIT), 2025.

## Research Experience

**Doctoral Researcher** — EURECOM, Biot, France Apr 2023 – Jul 2026

- Proved **information-theoretic converse bounds** on the communication load of distributed computing and matched them with explicit schemes, establishing order-optimality and exact asymptotic optimality over broad parameter regimes.
- Designed **universal (task-blind) data and task allocations**, built on new clique- and hypergraph-decomposition constructions (Interweaved Cliques; Hamiltonian decompositions of complete $k$-uniform hypergraphs), with guarantees that hold for every set of requested computations.
- Characterized **fundamental limits under random sampling** of the requested computations (random hypergraphs), combining counting arguments with **concentration inequalities and high-dimensional probability**.

- Planned, carried out, and published this program largely independently; implemented constructions and numerical evaluations in **Python**.

**Ongoing Research** — EURECOM & collaborators 2026 – present

- **Communication vs. load balancing in mixture-of-experts models:** expert placement as a distributed computing problem with routing-driven random demands; converse bounds and placements learned from observed routing data.
- **Beyond independent sampling:** extending the limits above to **correlated** demands, with counting and min-entropy converse arguments.
- **Certified resource budgets:** distribution-free, finite-sample guarantees for system budgets calibrated from data.

**Undergraduate Researcher — Graph Learning & Large-Scale Graph Algorithms** — Sharif University of Technology & Zarebin Jun – Dec 2022

- Investigated **graph Laplacian learning** to infer graph structure from multivariate (stock-market) data; formulated and solved the underlying convex programs.
- Developed scalable **PageRank-style algorithms on massive graphs** with Apache Spark / GraphX.

**Undergraduate Thesis – Statistical Signal Processing** — Sharif University of Technology Sep 2021 – Mar 2022
Adaptive and Kalman filtering pipelines for ECG denoising and source separation.

**Research Intern – Applied Deep Learning** — Sharif University of Technology Jun – Aug 2019
Leukocyte-nuclei segmentation in blood-smear images via color transforms and deep networks; peer-reviewed publication.

## Teaching Experience

Teaching Assistant – EURECOM & Sharif University of Technology (2019 – 2023)

Digital Communications • Probability & Statistics • Digital Signal Processing • undergraduate tutorials in Engineering Mathematics and Graph Signal Processing. Duties included problem classes, problem sets and solutions, office hours, and grading.

## Awards & Distinctions

- **Rank 7** / **180,000+**, Iranian National University Entrance Examination (B.Sc. level).
- Held a draw against GM Kotronias at the ISIT 2024 chess event; featured on EURECOM news and the ISIT 2024 website.

## Technical Skills & Languages

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|---|---|
| **Theory** | Information theory (converse bounds, information measures), network and distributed computing, learning theory, minimax lower bounds (Fano, Le Cam), high-dimensional probability, concentration inequalities, conformal prediction and risk control, convex and online optimization, graph & hypergraph theory, combinatorial designs |
| **ML systems** | Mixture-of-experts routing and expert parallelism, Transformers, diffusion models, RLHF/DPO |
| **Programming** | Python (advanced), MATLAB, LaTeX (advanced); C/C++ (intermediate) |
| **Frameworks** | PyTorch, TensorFlow, NumPy/SciPy, Apache Spark / GraphX, Git, Linux |
| **Languages** | English • French (B1, actively improving) • Persian (native) |

## Referees

- **Prof. Petros Elia** *PhD supervisor*
  Professor, Communication Systems Department, EURECOM, Sophia Antipolis, France
  petros.elia@eurecom.fr
- **Dr. Gholamali Aminian** *Research collaborator*
  Research Associate, The Alan Turing Institute, London, UK
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